DPP Solutions

```
try:
    numbers = [5, 10, 15]

    index = 2

    print(numbers[index])

except IndexError:
    print("Wrong Index")

else:
    print("Successful")

finally:
    print("Done")
```

```
15
Successful
Done
```

```
try:
    age = int("Twenty")

    print(age)

except ValueError:
    print("Invalid Age")

finally:
    print("End")
```

```
Invalid Age
End
```

```
try:
    student = {
        "name": "Rahul",
        "marks": 90
    }

    print(student["age"])

except KeyError:
    print("Key Missing")

print("Program End")
```

```
Key Missing
Program End
```

```
try:
    marks = 120

    if marks > 100:
        raise ValueError("Marks Invalid")

    print(marks)

except ValueError as e:
    print(e)

print("Completed")
```

```
Marks Invalid
Completed
```

```
try:
    marks = 90

    if marks > 100:
        raise ValueError("Marks Invalid")

    print(marks)

except ValueError as e:
    print(e)

print("Completed")
```

```
90
Completed
```

```
try:
    num = int("25")

    print(100 / num)

except ValueError:
    print("Value Error")

except ZeroDivisionError:
    print("Division Error")

else:
    print("Calculation Done")
```

```
4.0
Calculation Done
```

```
try:
    numbers = [2, 4, 6]

    print(numbers[5])

except IndexError:
    print("Index Error")

except Exception:
    print("General Error")

finally:
    print("Finished")
```

```
Index Error
Finished
```

```
try:
    value = int("50")

    print(value)

except Exception:
    print("Error")

else:
    print("Success")

finally:
    print("Done")
```

```
50
Success
Done
```

Case-Study

```
students = [
    {
        "roll":101,
```

```
        "name":"Rahul",
        "age":20,
        "course":"Python",
        "marks":85
    },
    {
        "roll":102,
        "name":"Neha",
        "age":19,
        "course":"SQL",
        "marks":92
    },
    {
        "roll":103,
        "name":"Amit",
        "age":22,
        "course":"Python",
        "marks":74
    },
    {
        "roll":104,
        "name":"Priya",
        "age":21,
        "course":"AI",
        "marks":96
    },
    {
        "roll":105,
        "name":"Rohan",
        "age":20,
        "course":"Python",
        "marks":67
    }
]
```

```
students
```

```
[{'roll': 101, 'name': 'Rahul', 'age': 20, 'course': 'Python', 'marks': 85},
 {'roll': 102, 'name': 'Neha', 'age': 19, 'course': 'SQL', 'marks': 92},
 {'roll': 103, 'name': 'Amit', 'age': 22, 'course': 'Python', 'marks': 74},
 {'roll': 104, 'name': 'Priya', 'age': 21, 'course': 'AI', 'marks': 96},
 {'roll': 105, 'name': 'Rohan', 'age': 20, 'course': 'Python', 'marks': 67}]
```

```
type(students)
```

```
list
```

```
type(students[0])
```

```
dict
```

```
# Display the details of every student stored in the institute.
```

```
for student in students:
  print(student)
```

```
{'roll': 101, 'name': 'Rahul', 'age': 20, 'course': 'Python', 'marks': 85}
{'roll': 102, 'name': 'Neha', 'age': 19, 'course': 'SQL', 'marks': 92}
{'roll': 103, 'name': 'Amit', 'age': 22, 'course': 'Python', 'marks': 74}
{'roll': 104, 'name': 'Priya', 'age': 21, 'course': 'AI', 'marks': 96}
{'roll': 105, 'name': 'Rohan', 'age': 20, 'course': 'Python', 'marks': 67}
```

```
# Print only the names of all students.


for student in students:
  print(student["name"])
```

```
Rahul
Neha
Amit
Priya
Rohan
```

```
# Calculate the total marks obtained by all students.

total = 0

for student in students:
  print(student["marks"])
  total += student["marks"]

print(total)
```

```
85
92
74
96
67
414
```

```
# Find Average Marks of the class


total = 0
n = len(students)
for student in students:
  print(student["marks"])
  total += student["marks"]

print(total/n)
```

```
85
92
74
96
67
82.8
```

```
# Count Passed Students

# Pass Marks = 40


count = 0
for student in students:
  if student["marks"] >= 40:
    count += 1

print(count)
```

```
5
```

```
# Count Passed Students

# Pass Marks = 90


count = 0
for student in students:
  if student["marks"] >= 90:
    count += 1

print(count)
```

```
2
```

```
Write a function to Print Grade of Every Student

Rules

90+

A

75+

B

60+
```

```
C

Else

D
```

```
def calc_grade(marks):
  if marks > 90:
    grade = "A"
  elif marks > 75:
    grade = "B"
  elif marks > 60:
    grade = "C"
  else:
    grade = "D"

  return grade
```

```
for student in students:
  print(student["marks"], calc_grade(student["marks"]))
```

```
85 B
92 A
74 C
96 A
67 C
```

```
# Create a function to Search Student by their Roll Number

def search(roll_no):
  for student in students:
    if student["roll"] == roll_no:
      print(student)
      return "Record Found"

  return "Not Found"
```

```
roll = int(input())

ans = search(roll)
print(ans)
```

```
101
{'roll': 101, 'name': 'Rahul', 'age': 20, 'course': 'Python', 'marks': 85}
Record Found
```

```
roll = int(input())

ans = search(roll)
print(ans)
```

```
200
Not Found
```

```
students
```

```
[{'roll': 101, 'name': 'Rahul', 'age': 20, 'course': 'Python', 'marks': 85},
 {'roll': 102, 'name': 'Neha', 'age': 19, 'course': 'SQL', 'marks': 92},
 {'roll': 103, 'name': 'Amit', 'age': 22, 'course': 'Python', 'marks': 74},
 {'roll': 104, 'name': 'Priya', 'age': 21, 'course': 'AI', 'marks': 96},
 {'roll': 105, 'name': 'Rohan', 'age': 20, 'course': 'Python', 'marks': 67}]
```

```
# Search Student by Name

# Ignore case while searching.

name = input("Enter name:")

for student in students:
  if student["name"].lower() == name.lower():
    print(student)
```

```
Enter name:Rahul
{'roll': 101, 'name': 'Rahul', 'age': 20, 'course': 'Python', 'marks': 85}
```

```
# Search Student by Name

# Ignore case while searching.

name = input("Enter name:")

for student in students:
  if student["name"].lower() == name.lower():
    print(student)
```

```
Enter name:rahul
{'roll': 101, 'name': 'Rahul', 'age': 20, 'course': 'Python', 'marks': 85}
```

```
# Search Student by Name

# Ignore case while searching.

name = input("Enter name:")

for student in students:
  if student["name"].lower() == name.lower():
    print(student)
```

```
Enter name:aj
```

```
# Write a function to print the topper of the class.

def find_topper(students):
  mx_marks = 0
  topper = {}

  for student in students:
    if student["marks"] > mx_marks:
      mx_marks = student["marks"]
      topper = student


  return topper
```

```
ans = find_topper(students)
print(ans)
```

```
{'roll': 104, 'name': 'Priya', 'age': 21, 'course': 'AI', 'marks': 96}
```

```
# Count Students in Each Course

students
```

```
[{'roll': 101, 'name': 'Rahul', 'age': 20, 'course': 'Python', 'marks': 85},
 {'roll': 102, 'name': 'Neha', 'age': 19, 'course': 'SQL', 'marks': 92},
 {'roll': 103, 'name': 'Amit', 'age': 22, 'course': 'Python', 'marks': 74},
 {'roll': 104, 'name': 'Priya', 'age': 21, 'course': 'AI', 'marks': 96},
 {'roll': 105, 'name': 'Rohan', 'age': 20, 'course': 'Python', 'marks': 67}]
```

```
# expected op:
# python - 3
# sql - 1
# ai - 1
```

```
courses = {}
for student in students:
  if student["course"] not in courses:
    courses[student["course"]] = 1
  else:
    courses[student["course"]] += 1

print(courses)
```

```
{'Python': 3, 'SQL': 1, 'AI': 1}
```

```
# Remove Duplicate Courses

courses = []

for student in students:
  courses.append(student["course"])

print(courses)
s = set(courses)
print(s)
```

```
['Python', 'SQL', 'Python', 'AI', 'Python']
{'SQL', 'Python', 'AI'}
```

```
students
```

```
[{'roll': 101, 'name': 'Rahul', 'age': 20, 'course': 'Python', 'marks': 85},
 {'roll': 102, 'name': 'Neha', 'age': 19, 'course': 'SQL', 'marks': 92},
 {'roll': 103, 'name': 'Amit', 'age': 22, 'course': 'Python', 'marks': 74},
 {'roll': 104, 'name': 'Priya', 'age': 21, 'course': 'AI', 'marks': 96},
 {'roll': 105, 'name': 'Rohan', 'age': 20, 'course': 'Python', 'marks': 67}]
```

```
# Create Student Class (OOP) in the above format

class Student:

  def __init__(self, roll, name, age, course, marks):
    self.roll = roll
    self.name = name
    self.age = age
    self.course = course
    self.marks = marks

  def show_student(self):
    print(self.roll, self.name, self.age, self.course, self.marks)
```

```
s1 = Student(106, "MJ", 30, "Dance", 100)
```

```
s1.show_student()
```

```
106 MJ 30 Dance 100
```

```
students
```

```
[{'roll': 101, 'name': 'Rahul', 'age': 20, 'course': 'Python', 'marks': 85},
 {'roll': 102, 'name': 'Neha', 'age': 19, 'course': 'SQL', 'marks': 92},
 {'roll': 103, 'name': 'Amit', 'age': 22, 'course': 'Python', 'marks': 74},
 {'roll': 104, 'name': 'Priya', 'age': 21, 'course': 'AI', 'marks': 96},
 {'roll': 105, 'name': 'Rohan', 'age': 20, 'course': 'Python', 'marks': 67}]
```

```
Build Complete Student Management Menu

2 Display Students

3 Search Student

4 Find Topper

5 Calculate Average
```

```
6 Exit
```

```
def search(roll_no):
  for student in students:
    if student["roll"] == roll_no:
      print(student)
      return "Record Found"

  return "Not Found"


def find_topper(students):
  mx_marks = 0
  topper = {}

  for student in students:
    if student["marks"] > mx_marks:
      mx_marks = student["marks"]
      topper = student


  return topper
```

```
while True:

  choice = int(input("Enter your choice: "))

  if choice == 2:
      for student in students:
          print(student)

  elif choice == 3:
      roll = int(input("Enter student Roll no: "))
      print(search(roll))

  elif choice == 4:
      print(find_topper(students))

  elif choice == 5:
      total = 0
      for student in students:
          total += student["marks"]

      print("Average:", total / len(students))

  elif choice == 6:
      print("Thank You!")

  else:
      print("Invalid Choice")
      break
```

```
Enter your choice: 2
{'roll': 101, 'name': 'Rahul', 'age': 20, 'course': 'Python', 'marks': 85}
{'roll': 102, 'name': 'Neha', 'age': 19, 'course': 'SQL', 'marks': 92}
{'roll': 103, 'name': 'Amit', 'age': 22, 'course': 'Python', 'marks': 74}
{'roll': 104, 'name': 'Priya', 'age': 21, 'course': 'AI', 'marks': 96}
{'roll': 105, 'name': 'Rohan', 'age': 20, 'course': 'Python', 'marks': 67}
Enter your choice: 3
Enter student Roll no: 101
{'roll': 101, 'name': 'Rahul', 'age': 20, 'course': 'Python', 'marks': 85}
Record Found
Enter your choice: 4
{'roll': 104, 'name': 'Priya', 'age': 21, 'course': 'AI', 'marks': 96}
Enter your choice: 5
Average: 82.8
Enter your choice: 6
Thank You!
Enter your choice: 8
Invalid Choice
```

```
Start coding or generate with AI.
```